

Scientific Programming with Python

Assignment: Soil Contamination by Factories

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July 13, 2026

Goal The goal of this assignment is to apply the knowledge of scientific computing acquired during this course and to learn a bit about soil contamination (e.g., from heavy metals such as lead and mercury) [1]. Specifically, you will analyze soil sample data collected around an industrial complex by clustering the data into distinct contamination zones, allowing you to identify the extent of the pollution caused by each building. You will also optimize parameters of a simple “plume” (i.e., the dispersion pattern) model for one of the buildings to estimate its contamination source and its concentration.

Problem and Input Data Topsoil contamination is a serious environmental concern that impacts plant-based food production, animals, and human health [2]. This is particularly important in regions near urban development. To illustrate this, we have the following scenario - a company has an industrial complex with the three buildings shown in Figure 1. As an environmental scientist, you have collected soil samples around this complex, and you need to determine which building generates the most contamination and identify the exact source.

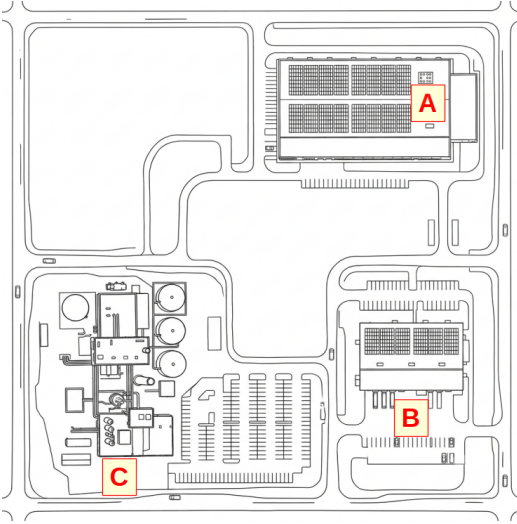


Figure 1. Three factories (i.e., A, B and C) in a 6.0 by 6.0 kilometer industrial complex. Source: Gemini generated.

The contamination’s dispersion pattern can be modeled by the following equation, which has five adjustable parameters (i.e., A , B , x_0 , y_0 , and L):

$$C(x, y) = B + A \exp\left(-\frac{r}{L}\right), \quad (1)$$

where $r = \sqrt{(x - x_0)^2 + (y - y_0)^2}$

where

- $C(x, y)$: predicted contaminant concentration at location (x, y) (e.g., milligrams/kilogram $\rightarrow$ milligrams of contaminant per kilogram of soil).
- A : plume amplitude (mg/kg).
- B : background concentration level (mg/kg).
- L : the decay length (meters) that controls how quickly the concentration decreases with distance (larger L = longer spread). This is basically the distance that the pollution travels from its source.
- r : distance (meters) from the contamination source to the point (x, y) .
- x, y : coordinates of a soil sample (meters).
- (x_0, y_0) : coordinates of the contamination source (meters).

The data file contains the latitude (i.e., x) and longitude (i.e., y) values – *relative to the industrial complex’s center* – and the contaminant concentration (i.e., $C(x, y)$).

Assignment Tasks

Task 1: Data Preparation Load the `soil_sample.csv` data as a DataFrame.

Task 2: Clustering and Visualization

- (a) Cluster the data by its latitude and longitude values using scikit-learn's K-means algorithm.

Hint: I used a `random_state=0`.

- (b) Add the resulting cluster label to Task 1's DataFrame.
(c) Create a scatter plot of the resulting clusters.

Task 3: Model Parameter Optimization for Building A

- (a) Create a user-defined function that encodes Equation 1.

(b) Optimize the five adjustable parameters of Equation 1 using SciPy's `curve_fit` for Cluster that corresponds to Building A in Figure 1. Use the following as reasonable initial guesses:

$$\begin{array}{lll} A = 1000 & X_0 = 0 & L = 500 \\ B = 0 & Y_0 = 1000 & \end{array}$$

Hint: In my solution, Building A corresponds to the cluster with `label = 1`.

Task 4: Predict Building A's Contaminant Concentration and Visualize

- (a) Using the optimized parameters from Task 3, predict Building A's contaminant concentration using Equation 1.
(b) Create a scatter plot of the real contaminant concentration data (i.e., from the CSV file) versus the predicted concentration.

Task 5: Briefly Comment on Your Notebook's Results

Allowed Python3 [3, 4] functions & libraries/modules

- All built-in functions
- The matplotlib [5], NumPy [6] Pandas [7] libraries
- The SciPy [8] and sklearn [9] libraries
- The typing library

Assignment Due Turn in your solution as a Jupyter-notebook to **LEA** by **July 27th, 2026 at 13:30**.

Note 1 : Please include your **Name and SciPro ID** at the **top** of your notebook.

Note 2 : You **do not** need to consider **significant figures**.

References

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